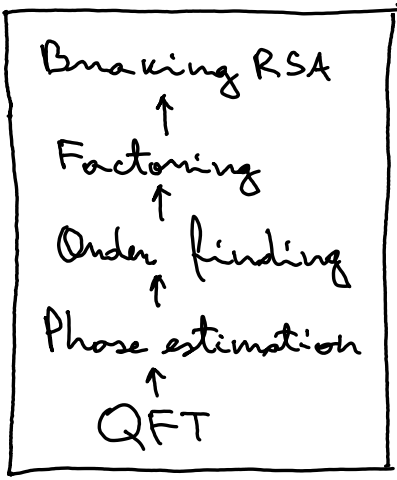
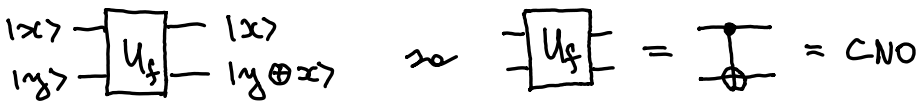
Lecture 10

This and the following lecture are dedicated to deriving Shor's algorithm for factoring integers. In this lecture we will cover quantum phase estimation (QPE) and the quantum Fourier transform (QFT). The algorithm for order finding and factoring will be covered in the next lecture.

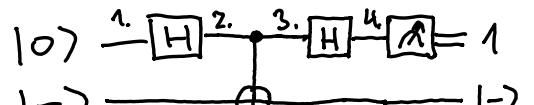


Let's recall Deutsch's algorithm and the phase kick-back trick for converting bit oracles to phase oracles.

Let $f:\{0,1\}\to\{0,1\}$ and $U_f|x,y\rangle = |x, y\oplus f(x)\rangle$, for all $x,y\in\{0,1\}$, be the bit oracle for f. Deutsch's algorithm determines $f(0)\oplus f(1)$ or whether f is constant or balanced. For concreteness, let's take $f(x)=x$, so $f(0)\oplus f(1) = 0\oplus 1 = 1$, i.e., f is balanced. For this choice of f, $U_f|x,y\rangle = |x, y\oplus x\rangle$, i.e.,

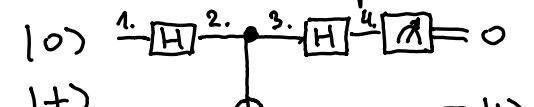


Since $f(0)\oplus f(1)=1$, Deutsch's algorithm outputs 1:



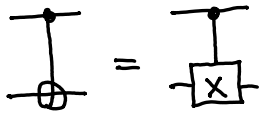
1. $|0\rangle|-\rangle$
2. $|+\rangle|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle|-\rangle + |1\rangle|-\rangle)$
3. $\frac{1}{\sqrt{2}}(|0\rangle|-\rangle - |1\rangle|-\rangle) = |-\rangle|-\rangle$
4. $|1\rangle|-\rangle$ ← because $X|-\rangle = -|-\rangle$

What if the second qubit is $|+\rangle$ instead of $|-\rangle$?



1. $|0\rangle|+\rangle$
2. $|+\rangle|+\rangle$
3. $|+\rangle|+\rangle$
4. $|0\rangle|+\rangle$

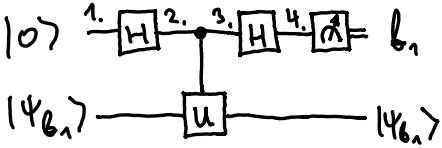
Here is how we can think about this. Note that



and $\begin{cases} X|+\rangle = |+\rangle \\ X|-\rangle = -|-\rangle \end{cases}$

where $|+\rangle$ and $|-\rangle$ are the eigenvectors of X. Given an eigenvector of X as a quantum state, Deutsch's algorithm determines which eigenvector it is. Equivalently, it determines the corresponding eigenvalue – if the measurement outcome is $b_1 \in \{0,1\}$ then the eigenvalue is $(-1)^{b_1}$ and the eigenstate was $|(-1)^{b_1}\rangle$. Note that this procedure preserves the eigenstate – indeed the final state is $|b_1\rangle|(-1)^{b_1}\rangle$. This is the same as measuring an observable on one of its eigenvectors!

This idea can be generalized to an arbitrary basis. Let $V \in U(2)$ be an arbitrary unitary basis change, and let $U = VXV^\dagger$ be X in a different basis. Since $X = |+\rangle\langle+| - |-\rangle\langle-| = |-\rangle\langle+| + |+\rangle\langle-|$, $U = |\psi_0\rangle\langle\psi_0| - |\psi_1\rangle\langle\psi_1|$ where $|\psi_0\rangle = V|+\rangle$ and $|\psi_1\rangle = V|-\rangle$ are the eigenvectors of U. We can generalize the algorithm as follows: for any $b_1 \in \{0,1\}$



1. $|0\rangle|\psi_{b_1}\rangle$
2. $|+\rangle|\psi_{b_1}\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle|\psi_{b_1}\rangle + |1\rangle|\psi_{b_1}\rangle\right)$
3. $\frac{1}{\sqrt{2}}\left(|0\rangle|\psi_{b_1}\rangle + |1\rangle(-1)^{b_1}|\psi_{b_1}\rangle\right)$
$= \frac{1}{\sqrt{2}}\left(|0\rangle + (-1)^{b_1}|1\rangle\right)|\psi_{b_1}\rangle = |(-1)^{b_1}\rangle|\psi_{b_1}\rangle$

($U|\psi_{b_1}\rangle = (-1)^{b_1}|\psi_{b_1}\rangle$)

4. $|b_1\rangle|\psi_{b_1}\rangle$

Again, we determine the eigenvalue of $|\psi_{b_1}\rangle$ without destroying the state itself (it belongs to the basis in which we measure it).

This idea can be generalized beyond ± 1 eigenvalues to an arbitrary eigenvalue $e^{i\varphi}$, $\varphi \in [0, 2\pi)$.

Quantum phase estimation

Let $U \in U(d)$ and $|\psi\rangle \in \mathbb{C}^d$ be an eigenvector of U:

$$U|\psi\rangle = e^{2\pi i \cdot b}|\psi\rangle, \text{ for some } b \in [0,1).$$

Problem: Given access to a subroutine for controlled-U and one copy of $|\psi\rangle$, determine b (i.e., the phase / the eigenvalue of $|\psi\rangle$).

We already solved the case when $e^{2\pi i b} \in \{1,-1\}$, so $b \in \{0, 1/2\}$, or in binary $b \in \{0.0, 0.1\}$. In other words, $b = 0.b_1$ and we determined $b_1 \in \{0,1\}$. How about 2-bit accuracy? If $b = 0.b_1b_2$, how do we find $b_1, b_2 \in \{0,1\}$?

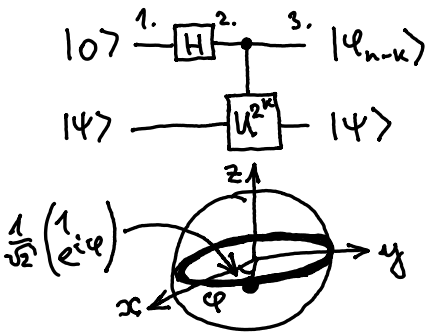
Note that $U^2|\psi\rangle = e^{2\pi i \cdot 2b}|\psi\rangle = e^{2\pi i \cdot 0.b_2}$ since $2b = 10_2 \cdot 0.b_1b_2 = b_1.b_2$ and $e^{2\pi i} = 1$. Hence, by using U^2 instead of U, we can determine b_2. But how about b_1?

Let's make the following general observation: if $b = 0.b_1b_2 \dots b_n$ then

$$U^{2^k}|\psi\rangle = e^{2\pi i \cdot 2^k \cdot b}|\psi\rangle$$

where $2^k \cdot b = \underbrace{10\dots0_2}_{k} \cdot 0.b_1b_2\dots b_n = \underbrace{b_1b_2\dots b_k}_{\text{integer part}}.\underbrace{b_{k+1}b_{k+2}\dots b_n}_{\text{fractional part with } n-k \text{ digits}}$

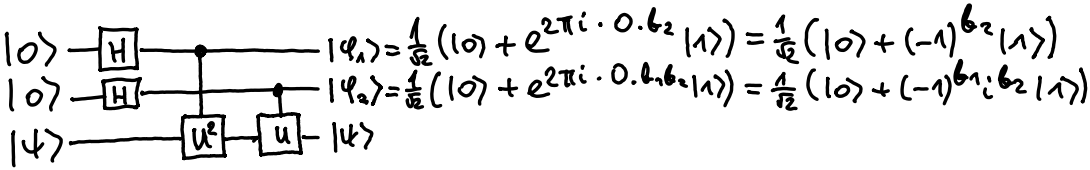
so $e^{2\pi i \cdot 2^k \cdot b} = e^{2\pi i \cdot 0.b_{k+1}\dots b_n}$.



1. $|0\rangle|\psi\rangle$
2. $|+\rangle|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle|\psi\rangle + |1\rangle|\psi\rangle)$
3. $\frac{1}{\sqrt{2}}(|0\rangle|\psi\rangle + |1\rangle e^{2\pi i \cdot 2^k \cdot b}|\psi\rangle)$

$$= \underbrace{\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i \cdot 0.b_{k+1}\dots b_n}|1\rangle)}_{|\psi_{n-k}\rangle}|\psi\rangle$$

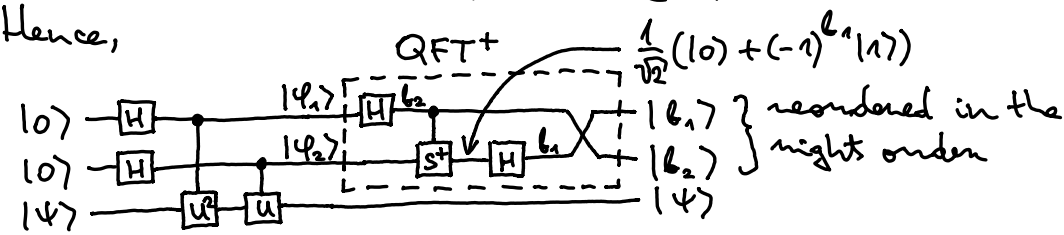
Let's see what happens if we produce both the state with $k=0$ and $k=1$ when $n=2$, and see how we can recover b_1 from this.



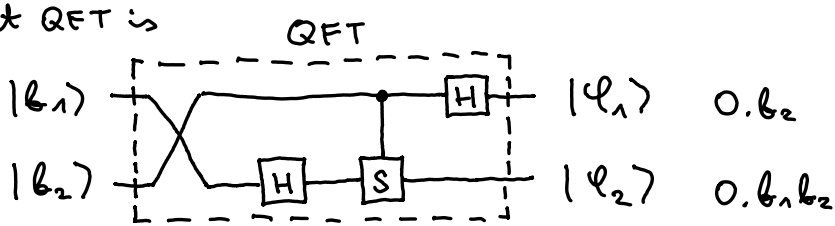
We can first recover b_2 the same way as before — by applying H on the first qubit. Once we have b_2, we can remove it from the second state as follows: let $S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$ and note that

$$S^+ = \begin{pmatrix} 1 & 0 \\ 0 & i^{-1} \end{pmatrix} \quad \text{so} \quad (S^+)^{b_2}|0\rangle = |0\rangle$$
$$(S^+)^{b_2}|1\rangle = (i^{-b_2})|1\rangle$$

Hence,



The <u>inverse</u> of the highlighted circuit is called the <u>quantum Fourier transform</u> (QFT). So the 2-qubit QFT is

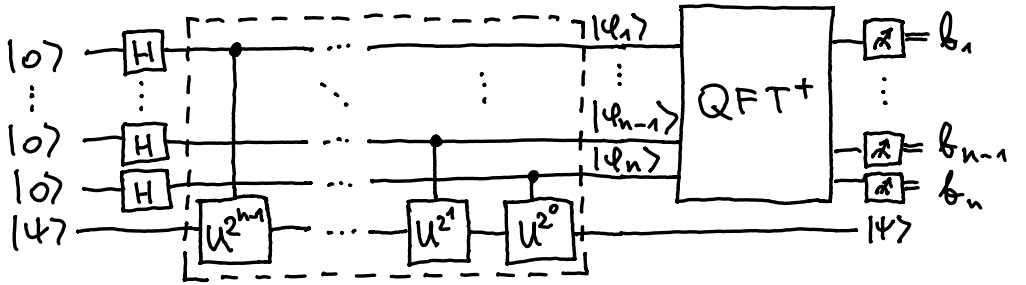


where $|\varphi_k\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i \cdot 2^{n-k} \cdot 0.b_{n-k+1}\ldots b_n}|1\rangle)$. More generally,

$$\text{QFT}\,|b_1 b_2 \ldots b_n\rangle = |\varphi_1\rangle \otimes |\varphi_2\rangle \otimes \ldots \otimes |\varphi_n\rangle.$$

and the circuit can be obtained recursively.

The general quantum phase estimation circuit for estimating the phase of an eigenvalue of U up to n binary digits of precision looks as follows:



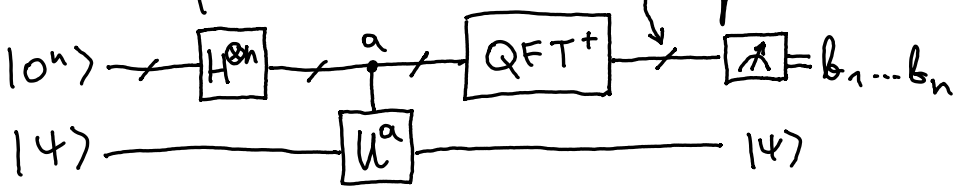
Here is how you can think of the highlighted part. If the first n qubits are in the computational basis state $|a_1 a_2 \dots a_n\rangle$, for some $a_i \in \{0,1\}$, the total power of U is

$$a_1 2^{n-1} + a_2 2^{n-2} + \dots + a_{n-1} 2^1 + a_n 2^0 = a$$

where $a \in \{0, \dots, 2^n - 1\}$ is the integer whose binary representation is $a_1 a_2 \dots a_n$. Hence this part implements the following operation:

$$\sum_{a=0}^{2^n-1} |a\rangle\langle a| \otimes U^a$$

We can abbreviate the quantum phase estimation circuit as follows:



This is somewhat reminiscent of Simon's algorithm, except for QFT^+ instead $H^{\otimes n}$ and a-controlled-U instead of the bit oracle U_f for some f.

Quantum Fourier transform

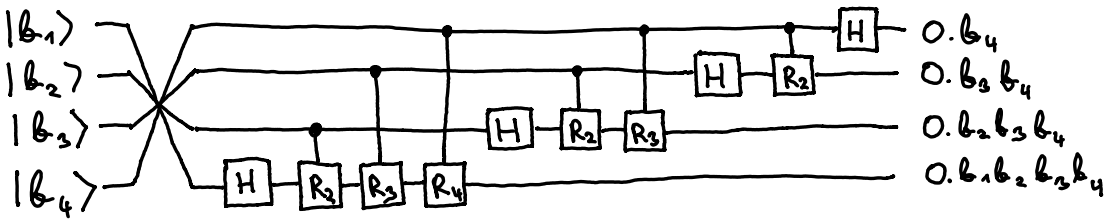
We defined the n-qubit quantum Fourier transform as

$$QFT\,|b_1, b_2, \dots, b_n\rangle = |\varphi_1\rangle \otimes |\varphi_2\rangle \otimes \dots \otimes |\varphi_n\rangle,$$

for all $b_1, b_2, \dots, b_n \in \{0,1\}$ where

$$|\varphi_{n-k}\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i \cdot 2^k \cdot \overbrace{0.b_{k+1}\dots b_n}} |1\rangle\right)$$

For $n=4$ the QFT circuit looks like this:



where R_k rotates around the z axis by angle $2\pi/2^k$:

$$R_k = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{pmatrix}$$

For example, $R_0 = I$, $R_1 = Z$, $R_2 = S$.

Discrete Fourier transform

Quantum Fourier transform is a special case of the discrete Fourier transform which exists for any matrix size $M \geq 1$, not just $M = 2^n$:

$$QFT(n) = D(2^n)$$

$D(M)$ is an $M \times M$ complex matrix.

The entries of $D(M)$ are as follows:

$$D_{jk} = \frac{1}{\sqrt{M}} \omega_M^{jk}, \quad j,k \in \{0, \dots, M-1\}$$

where $\omega_M = e^{2\pi i/M}$. Note that $\omega_M^M = 1$ and $\bar{\omega}_M = \omega_M^{-1}$. Clearly, $D^T = D$ since $D_{jk} = D_{kj}$, so $D^\dagger = \bar{D}$ and

$$(D^\dagger)_{jk} = \frac{1}{\sqrt{M}} \omega_M^{-jk}$$

We can easily write down the matrix for D:

$$D(M) = \frac{1}{\sqrt{M}} \begin{pmatrix} 1 & 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \omega^3 & \cdots & \omega^{M-1} \\ 1 & \omega^2 & \omega^4 & \omega^6 & \cdots & \omega^{2(M-1)} \\ 1 & \omega^3 & \omega^6 & \omega^9 & \cdots & \omega^{3(M-1)} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{M-1} & \omega^{2(M-1)} & \omega^{3(M-1)} & \cdots & \omega^{(M-1)^2} \end{pmatrix}$$

It is not obvious that D is unitary (exercise) and that $D(2^n)$ agrees with what is implemented by QFT(n) (exercise).

The intuition behind the discrete Fourier transform is as follows. The entries in each row of D correspond to points on the unit circle that run around it at increasingly high speeds. If $|y\rangle = D|x\rangle$ then each entry of $|y\rangle \in \mathbb{C}^M$,

$$y_j = \frac{1}{\sqrt{M}} \sum_{k=0}^{M-1} \omega^{jk} x_k,$$

is the inner product of $|x\rangle \in \mathbb{C}^M$ with a spiral of a certain period. This value is large if $|x\rangle$ oscillates with a similar period.

